

Asian HLA pangenome

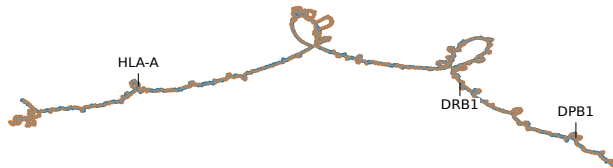
754 haplotype entries, including 462 Asian / Arab entries; 371 distinct donors



HPRC includes the 16 Japanese donors. Five donors overlap between projects; GRCh38 and CHM13 add two reference haplotypes.

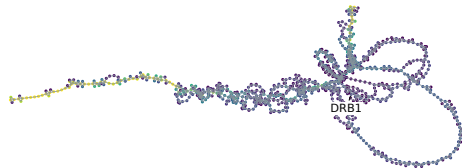
Source	Donor location	Samples
HPRC r2	Worldwide, incl. Japan	232
	Japanese subset (JPT)	16
APR	UAE	53
JaSaPaGe	Saudi Arabia / Japan	9 / 10
K-PanRef	Korea	14
CPC	China	58

MHC construction backbone



1,677 segments; blue: GRCh38, orange: added sequence
Read resource: 1000 Genomes ($n = 2504$)
BioHackathon Japan 2026

Class II / DRB bundle graph

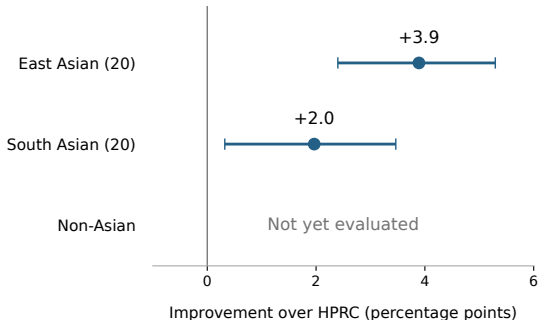


753 haplotypes; colour: bundle sharing (purple to yellow)

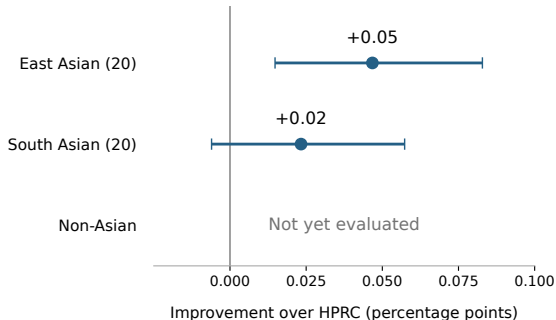
Variant recovery compared with HPRC

Larger gains for structural variants; small changes for SNVs

Structural variants



SNVs



Same reads and caller; expanded panel versus HPRC-only. Points: paired gains; bars: 95% intervals.

SV-bearing genotypes and shared SNV sites; horizontal scales differ. Test families excluded from panels; test assemblies remain in the graph.



Dgo**HLA**